CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 6: Triangles

Previous year board questions, topic-wise: 18 questions

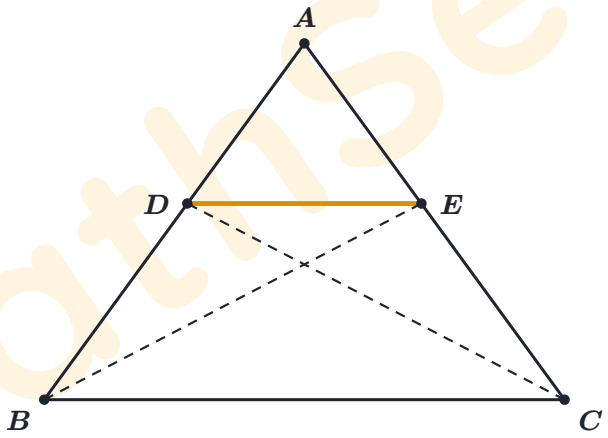
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Basic Proportionality Theorem

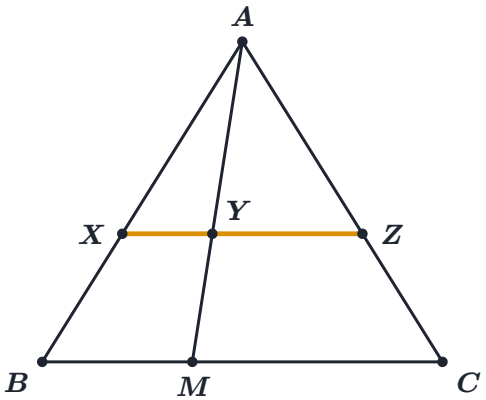
1. In $\triangle ABC$, D lies on AB and E lies on AC with $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find x. CBSE 2026, 2 marks

2. State the Basic Proportionality Theorem and prove it. CBSE 2026, 5 marks

3. Prove that a line drawn parallel to one side of a triangle, cutting the other two sides at distinct points, divides those two sides in the same ratio. CBSE 2024, 5 marks



4. In $\triangle ABC$, X lies on AB and Z on AC with $XZ \parallel BC$. M is a point on BC, and AM meets XZ at Y. If $AZ = 3$ cm, $ZC = 2$ cm, $BM = 3$ cm and $MC = 5$ cm, find XY. CBSE 2023, 2 marks



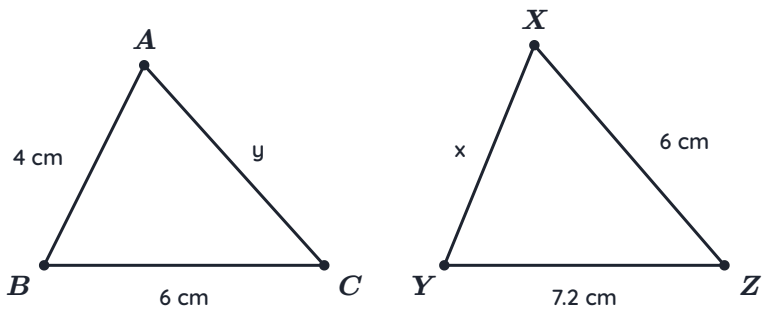
5. A line parallel to one side of a triangle meets the other two sides at distinct points. Prove that it divides those two sides in the same ratio. CBSE 2023, 5 marks

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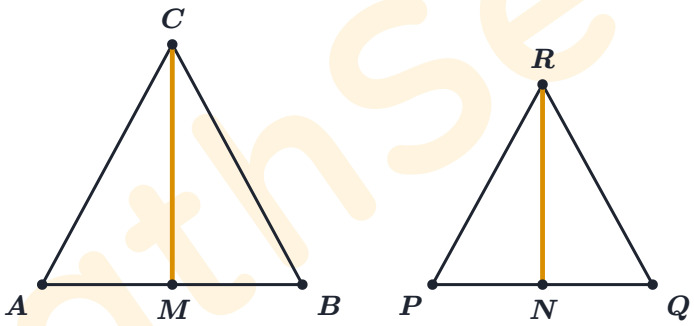
2 Criteria for similarity of triangles

6. $\triangle ABC \sim \triangle DEF$ with $2AB = DE$ and $BC = 8$ cm. Find EF. (a) 4 cm (b) 8 cm (c) 12 cm (d) 16 cm
CBSE 2026, 1 mark

7. $\triangle ABC \sim \triangle XYZ$ with $AB = 4$ cm, $BC = 6$ cm, $AC = y$ cm, $XY = x$ cm, $YZ = 7.2$ cm and $XZ = 6$ cm. Find x and y. CBSE 2026, 2 marks



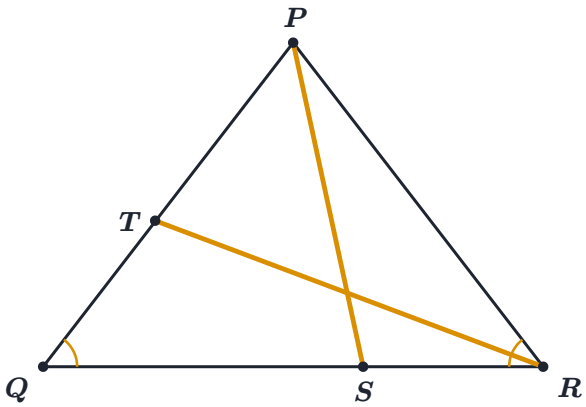
8. CM and RN are medians of $\triangle ABC$ and $\triangle PQR$, and $\triangle ABC \sim \triangle PQR$. Prove that (i) $\triangle AMC \sim \triangle PNR$ (ii) $\triangle CMB \sim \triangle RNQ$. CBSE 2026, 5 marks



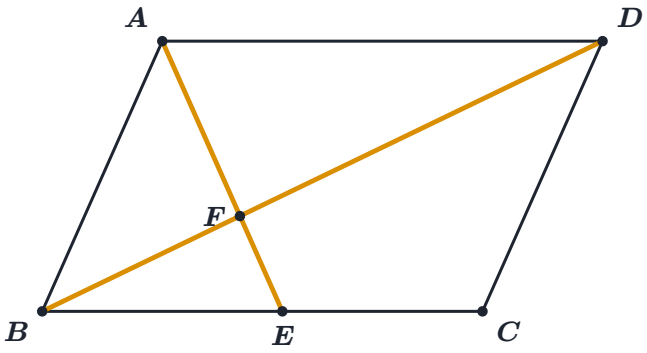
9. In $\triangle ABC$ and $\triangle DEF$, $\angle B = \angle E$, $\angle C = \angle F$ and $AB = 3DE$. Are the two triangles similar, congruent, both or neither? CBSE 2025, 1 mark

10. $\triangle ABC \sim \triangle PQR$ with $AB = 6$ cm, $BC = 4$ cm, $AC = 8$ cm and $PR = 6$ cm. Find $PQ + QR$.
CBSE 2025, 2 marks

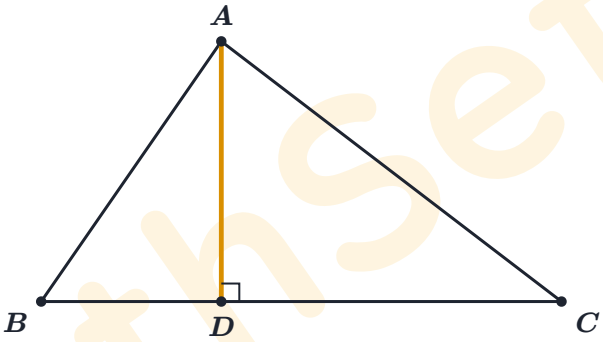
11. In $\triangle PQR$, S is a point on QR and T is a point on PQ. It is given that $\frac{QR}{QS} = \frac{QT}{PR}$ and $\angle PQR = \angle PRQ$. Prove that $\triangle PQS \sim \triangle TQR$. CBSE 2025, 2 marks



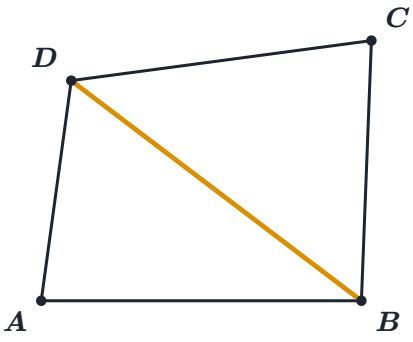
12. $ABCD$ is a parallelogram and E is a point on BC. The diagonal BD meets AE at F. Prove that $DF \times EF = FB \times FA$. CBSE 2025, 5 marks



13. In $\triangle ABC$, D is a point on BC with $AD \perp BC$ and $AD^2 = BD \times DC$. Prove that $\angle BAC = 90^\circ$. CBSE 2025, 5 marks

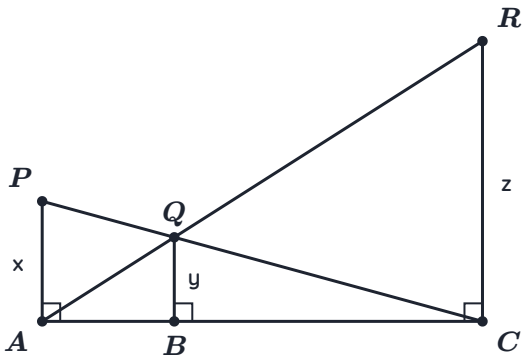


14. In quadrilateral $ABCD$, the diagonal BD bisects both $\angle B$ and $\angle D$. Prove that (i) $\triangle ABD \sim \triangle CBD$ (ii) $AB = BC$. CBSE 2024, 2 marks

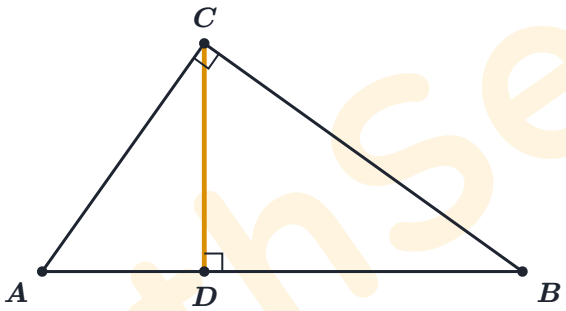


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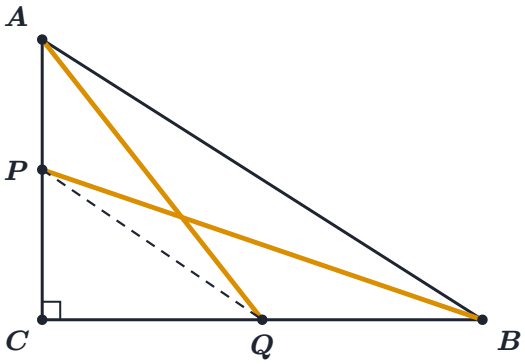
15. PA, QB and RC are all perpendicular to the line AC, where Q is the point in which PC and AR meet. If $AP = x$, $BQ = y$ and $CR = z$, prove that $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$. CBSE 2024, 5 marks



16. In $\triangle ABC$, $\angle ACB = 90^\circ$ and $CD \perp AB$. Prove that $CD^2 = BD \times AD$. CBSE 2019, 3 marks



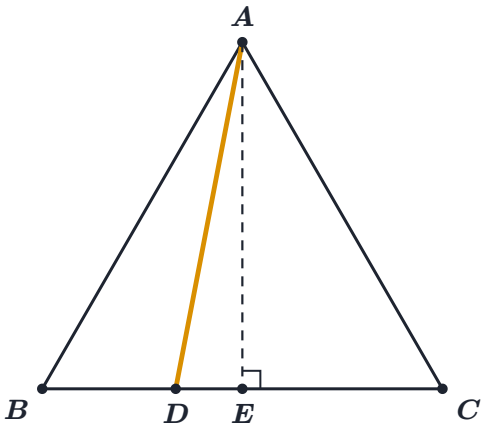
17. $\triangle ABC$ is right-angled at C. P is a point on CA and Q is a point on CB. Prove that $AQ^2 + BP^2 = AB^2 + PQ^2$. CBSE 2019, 3 marks



3 Right-angled triangles

18. ABC is an equilateral triangle and D is a point on BC with $BD = \frac{1}{3}BC$. Prove that $9AD^2 = 7AB^2$.

CBSE 2018, 4 marks



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